

There is no question that fishing is the major factor driving grouper stocks on the downward spiral, but those that have large spawning aggregations are most vulnerable to declines (Coleman et al. 1996; Asch and Erisman 2018; Sadovy de Mitcheson et al. 2020). Because it takes a long time for scientists to obtain needed life history information, fisheries-independent survey data, and catch history, grouper populations may be overfished long before data are even available for a stock assessment. Without formal stock assessments, general indicators of population status are based on catch trends. Very few grouper stocks that have spawning aggregations are managed sustainably. In a recent global analysis of the status of populations that form spawning aggregations, 45% were unknown, 33% were decreasing, and 5% were already gone (Figure 13.5). Only 12% had stable populations, and 5% were increasing.

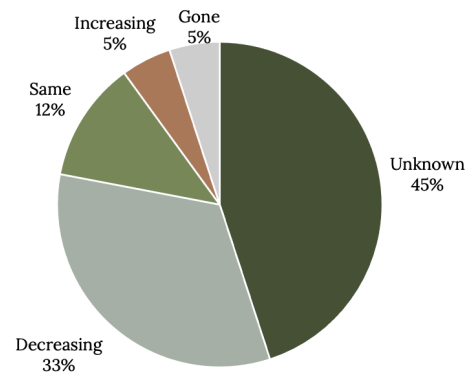


Figure 13.5: Current known status reflecting changes of exploited grouper aggregations globally, as noted by fisher interviews, monitoring, or underwater surveys (N = 509). [Long description](#).

Of the 167 species of grouper, 9.6% are vulnerable, 4.8% are near threatened, 1.2% are endangered, and 0.6% are critically endangered (Figure 13.6). The majority of species (68.9%) are classified as least concern and 15% are data deficient, with insufficient data for classification. The larger (>50 cm total length) and long-lived (>20 years) species of grouper that also had smaller geographic ranges were most likely to be endangered or critically endangered (Luiz et al. 2016). Market prices for grouper are escalating, and other lower-valued fish are often mislabeled or substituted.

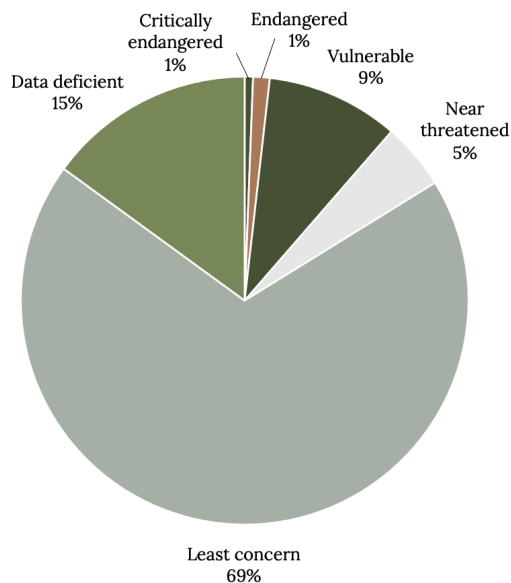


Figure 13.6: Categories of all grouper species (N = 167) according to the IUCN Red List (IUCN Red List Assessments, updated November 2018). [Long description](#).

To protect grouper from overfishing, many measures are being implemented, such as minimum and slot-size limits, recreational bag limits, commercial fishing quotas, gear and seasonal controls, marine protected areas, and limited entry (Rocklin et al. 2022). The effectiveness will depend on traits of the species and the local context. Regulations to prevent marketing of undersize fish will mitigate growth overfishing. Allowing smaller fish to reach maturity at least once before harvest will mitigate recruitment overfishing. Size-limit regulations focused on protecting spawning-size fish may be ineffective for deepwater recreational fishing. Grouper have a physoclistous (i.e., closed) swim bladder, making them particularly susceptible to ruptured swim bladders, bloating, stomach distention, and protruding eyes caused by rapid decompression when hauled to the surface (Brulé et al. 2015). The proportion of grouper with distended stomachs was 70% in one study of commercial hook-and-line fishing and as high as 95% for Red